



ReliefChain Engine

Disaster-Relief Supply-Chain & Priority Queue Dispatch Visualizer

Simulation Ready

Reset System

Live Network Simulation

Strategy Comparison & Benchmark

Dynamic Max-Heap Inspection

C Engine & Algorithmic Docs

Binary Max-Heap Visualizer

Select Strategy: Hybrid-Weighted



Array Representation & Sift Invariants

[0] Z5 69.8	[1] Z2 67.2	[2] Z0 51.9	[3] Z3 45.4
[4] Z1 36.8	[5] Z4 26.9		

- How the Binary Max-Heap Works
- Parent Index: $\text{parent} = (i - 1) / 2$
 - Left Child: $\text{left} = 2 * i + 1$
 - Right Child: $\text{right} = 2 * i + 2$
 - Dynamic Re-scoring: At each simulation tick, urgency spikes & waiting aging increase zone scores. Sift-up and sift-down maintain 'O(log N)' priority queue ordering.